Provenance-checked generation of class-compliant marine electrical design deliverables

Research concept note · Mashuk Elahi · 11 October 2026

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1. THE PROBLEM

A vessel's electrical design is not one document. It is a load list, a single-line diagram, a cable schedule, an alarm and monitoring point list, a drawing register, a set of supplier packages, and an order quantity summary — each authored at a different time, by a different party, and each quietly contradicting the others as revisions accumulate. On a 42 m newbuild the cable schedule alone runs to about 1,300 yard cables across 40 controlled drawings. Consistency between them is maintained today by an engineer reading everything again, and it fails in the same places every time: a cable resized on one sheet and not another, a supplier caption left beside a yard tag naming a different cable for the same run, a core count taken from a schedule rather than from the terminals the cores actually land on.

Large language models can now read these documents and reason about them. That makes them useful and dangerous in the same step, because a design value that is wrong in a cable schedule is not a bad sentence, it is an undersized conductor. The open question is not whether a model can draft a cable schedule. It is what structure around the model makes its output safe enough to issue under a classification society's eye — and how that structure is evidenced.

2. THE DEMONSTRATOR ALREADY RUNNING

This is not a plan from nothing. The following is in daily production use on one newbuild, built by me over the past year and used to issue real controlled documents:

- Deterministic generation. Every deliverable is rendered by a Python generator from one data set. A release is accepted only when two independent builds are byte-identical, so a build is reproducible evidence rather than a one-off artefact.
- A compliance linter as an issue gate, which must report zero errors before release. Its rules are real constraints: voltage grade against circuit voltage, minimum conductor size for a 0.6/1 kV family, flexible Class 5 construction on power multicores at and above 10 mm^2, fire-resistant duty on emergency-stop and fire circuits wherever they run.
- A provenance label on every engineering value: read from a named document and page, derived by a shown calculation, estimated, or not held. A value with neither source nor derivation cannot reach a deliverable, and a rule clause is citable only when its page has been read in the same working session.
- Independent machine verification by read-only agents that did not author the work, iterated to zero open findings. One full-set audit across roughly 650 pages returned 327 findings; the recurring families became permanent checks.
- A drawing application, H675 Draw: its own JSON drawing format, a 610-symbol library, SVG / DXF / PDF rendering, a 15-check verifier, and 40 tools through which an agent authors and marks up sheets. The verifier refuses tampered input; all 40 project drawings render and verify.
- Geometry extraction from arrangement DXF — trays, penetrations, frame lines — into a model from which cable routes are derived.

What the demonstrator does not yet have is a formal provenance schema, a published evaluation, or any engagement with a class society about what evidence would make such a pipeline acceptable. That is the research.

3. RESEARCH QUESTIONS

1. Where does the trust boundary belong? What classes of engineering decision can be delegated to a reading-and-proposing agent, which must remain with the signing engineer, and what observable property separates them?
2. What is the minimum provenance record for an engineering value such that a machine, not a human, can decide whether the value may be issued? Can that record be made complete enough to audit a deliverable without re-reading its sources?
3. Can rule-grounded citation be made reliable? Class and standards libraries are partial, previewed, scanned, or carry unmapped text layers. What retrieval and refusal behaviour keeps a clause citation honest when the page cannot be read?
4. What evidence would a classification society accept? Framed against Lloyd's Register SSC, AMSA NSCV and the IACS cyber-resilience unified requirements, what would an approvable AI-assisted design workflow have to demonstrate, and does reproducible generation plus independent verification supply it?

4. WORK PROGRAMME

Phase 1 — Formalise (months 1–6)

Extract the provenance model from the working system into a specification and schema. Define the status vocabulary, the admissibility rule for each status, and the machine check that enforces it. Survey the comparable literature: automated compliance checking in building information modelling, verified code generation, provenance standards for scientific data.

Phase 2 — Build the open demonstrator (months 4–12)

A reproducible pipeline, independent of any one yard's data, that takes a vessel load list through to a linted cable schedule, a single-line diagram mark-up and an alarm point list, with provenance carried end to end and a verifier that fails tampering. Released openly so the evaluation can be repeated.

Phase 3 — Evaluate (months 10–18)

Run the demonstrator against a real newbuild drawing set and measure what matters: defect classes caught by the linter against those caught only by human review; provenance completeness; reproducibility; and the residual class of errors that neither gate catches. Compare against the hand-authored baseline from the same project.

Phase 4 — Acceptability (months 16–24)

Put the evidence package to a classification society and a flag administration and record what is asked for that the pipeline does not yet produce. Write it up.

5. WHY THIS IS WORTH A SUPERVISOR'S TIME

- The subject is live. Yards are already putting language models near design documents, with no agreed structure for doing it safely and no published evaluation of what breaks.
- The demonstrator exists and has issued real documents under class scrutiny, so the work starts from a working system and a year of recorded failure modes rather than a literature gap.
- It bridges three fields that rarely meet: marine electrical engineering practice, automated compliance checking, and the engineering of trustworthy AI systems.
- It produces artefacts, not only papers: an open demonstrator, a provenance schema, and a recorded corpus of real design defects with the gate that would have caught each one.

6. WHAT I BRING

Sixteen years of marine, offshore and industrial electrical design and commissioning under ABS, BV, LR and DNV; a current hybrid-electric vessel newbuild; the toolchain above, written by me; a sole-authored systematic review on energy storage integration; and an MSc in digital transformation. I am seeking a tuition-free or funded research place

and a supervisor who wants this built properly rather than demonstrated once. I can work to a part-time or industry-embedded arrangement, with the newbuild programme as the evaluation case.

ANNEX A — OUTREACH EMAIL TO A PROSPECTIVE SUPERVISOR

Subject: Marine electrical design automation with machine-checked provenance — research place enquiry

Dear Professor <Name>,

I am a marine and offshore electrical engineer in Singapore, sixteen years in the field, currently delivering the hybrid-power architecture for a 42 m fast crew boat newbuild under Lloyd's Register and AMSA class. I am writing because I have built something I would like to turn into research, and your work on <specific area> is the closest fit I have found.

Over the past year I built a toolchain in which an AI agent reads the governing drawings and class rules and proposes engineering values, Python generators render every deliverable deterministically, and independent verifiers check the output before issue. It is in production use: a 1,300-cable schedule, a 40-drawing controlled set and an alarm point list that now stay consistent across revisions because every value carries a machine-checkable record of where it came from, and nothing is issued unless two builds agree byte for byte and the linter reports zero errors.

What I could not answer while building it is the research question: where the trust boundary belongs between an agent that reads and an engineer who signs, what the minimum provenance record for an engineering value is, and what evidence a classification society would need to accept such a workflow. I have attached a two-page concept note and my CV.

I am looking for a tuition-free or funded place and would be glad to discuss whether this fits your group. The newbuild programme can serve as the evaluation case, and the demonstrator can be released openly.

Thank you for your time.

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